between 3 and 5 percent in the coming year. There is less potential for both upside and downside with the bond, whereas the stock has much more potential for some great years, but also the potential risk of seriously dreary years. Hence, the standard deviation of expected returns informs investors of the amount of risk they're taking if they were to hold only that asset.

For a more holistic view, compare a portfolio with a standard deviation of returns of 4 percent to one that has a standard deviation of 8 percent. If both portfolios have the same expected return of 7 percent, it wouldn't be a prudent decision to invest in the portfolio with more volatility, as they both have the same expected return. Taking on a higher level of risk has no benefit in this light, and if a portfolio is unwisely constructed, investors can end up taking on more risk than they're compensated for.

Sharpe Ratio

Similar to the concepts behind MPT, the Sharpe ratio was also created by a Nobel Prize winner, William F. Sharpe. The Sharpe ratio differs from the standard deviation of returns in that it calibrates returns per the unit of risk taken. The ratio divides the average expected return of an asset (minus the risk-free rate) by its standard deviation of returns. For example, if the expected return is 8 percent, and the standard deviation of returns is 5 percent, then its Sharpe ratio is 1.6. The higher the Sharpe ratio, the better an asset is compensating an investor